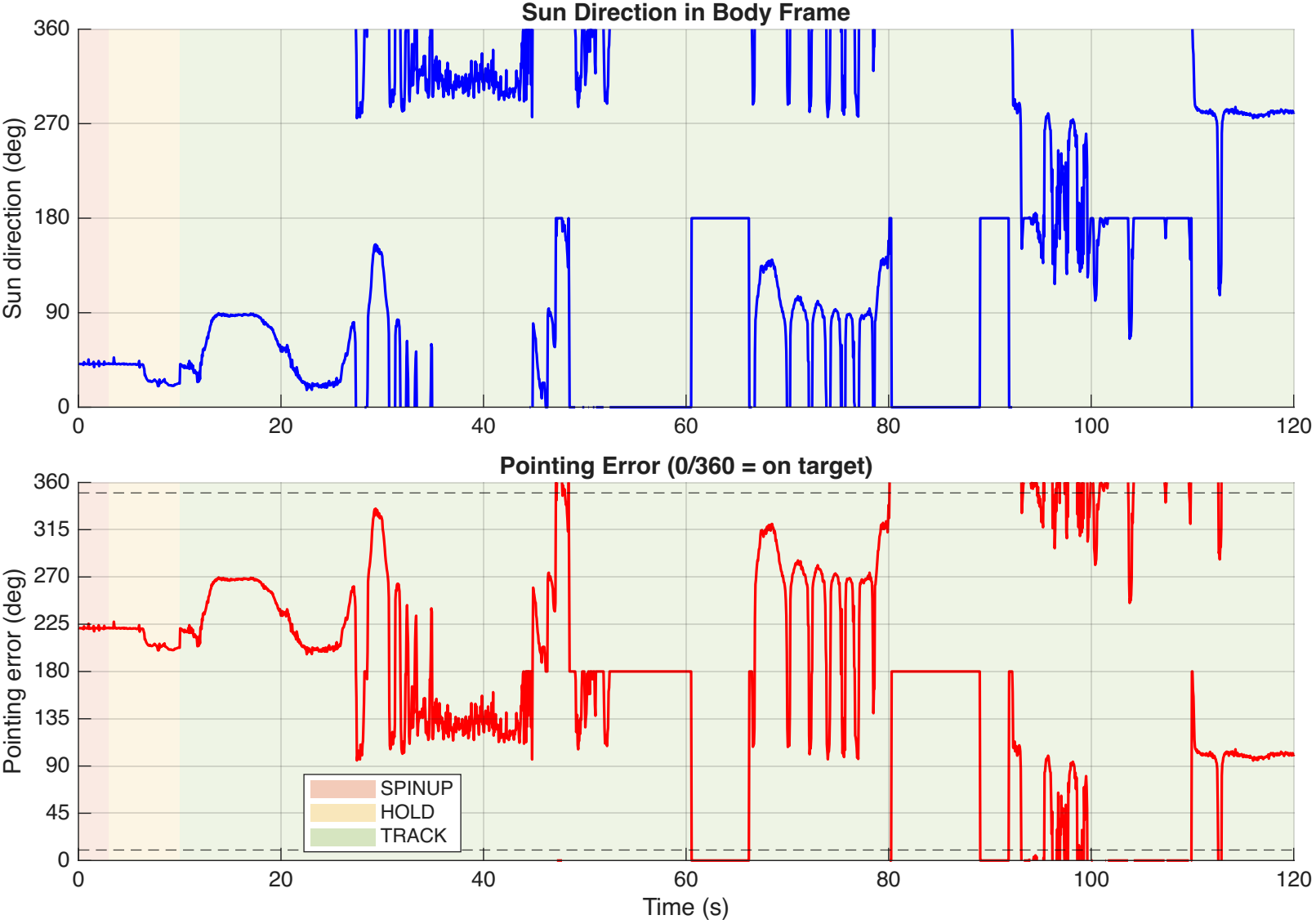
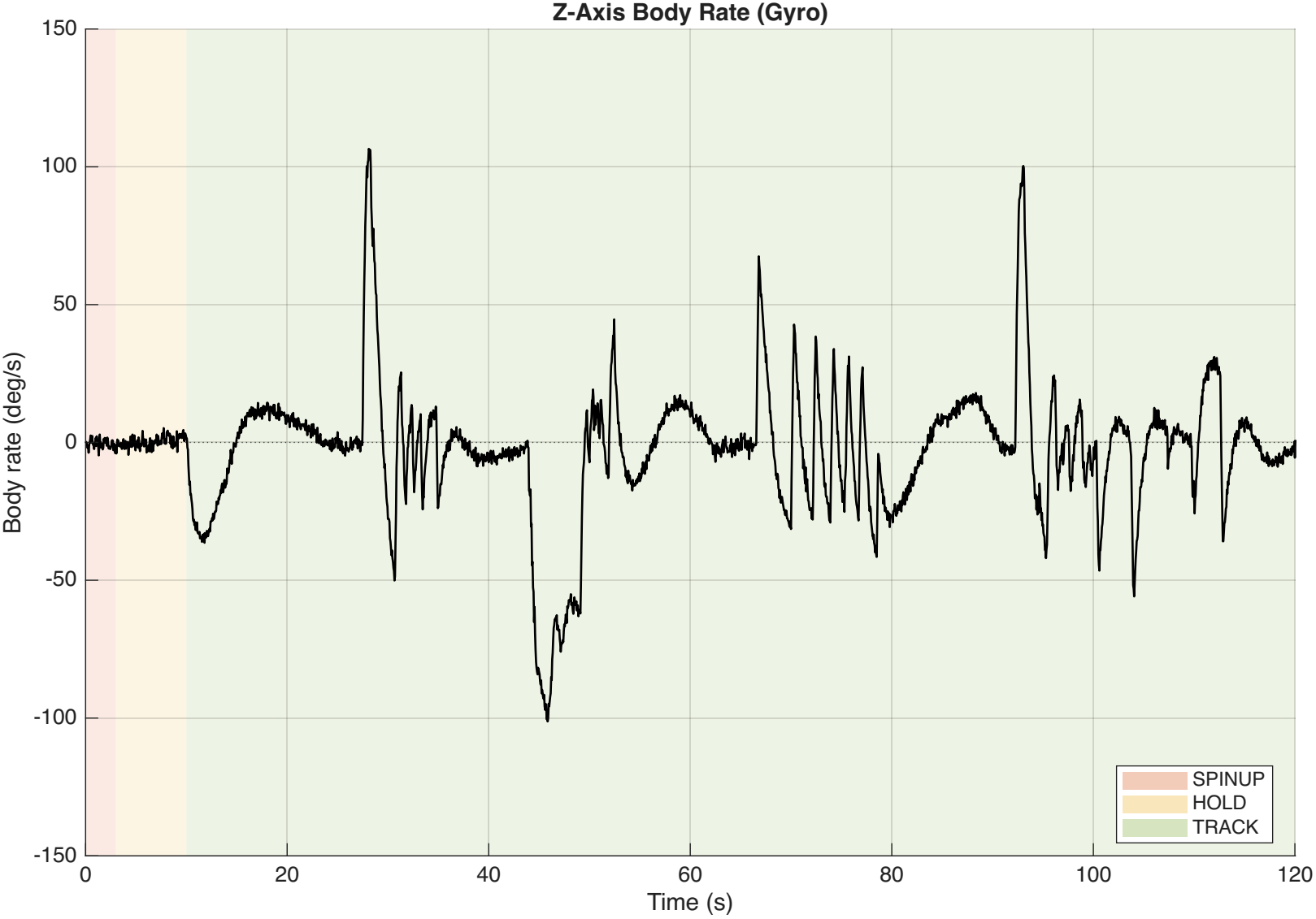
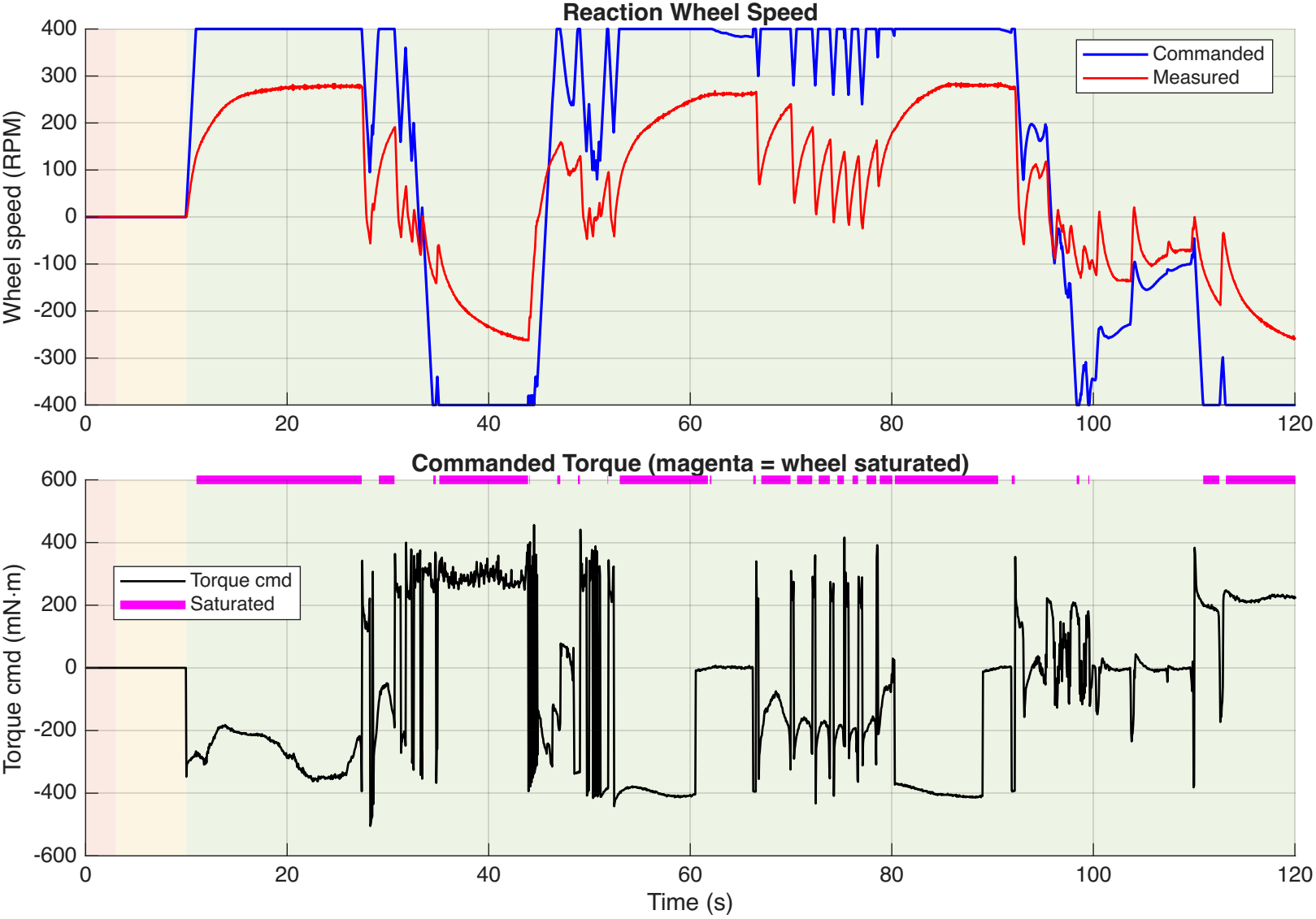
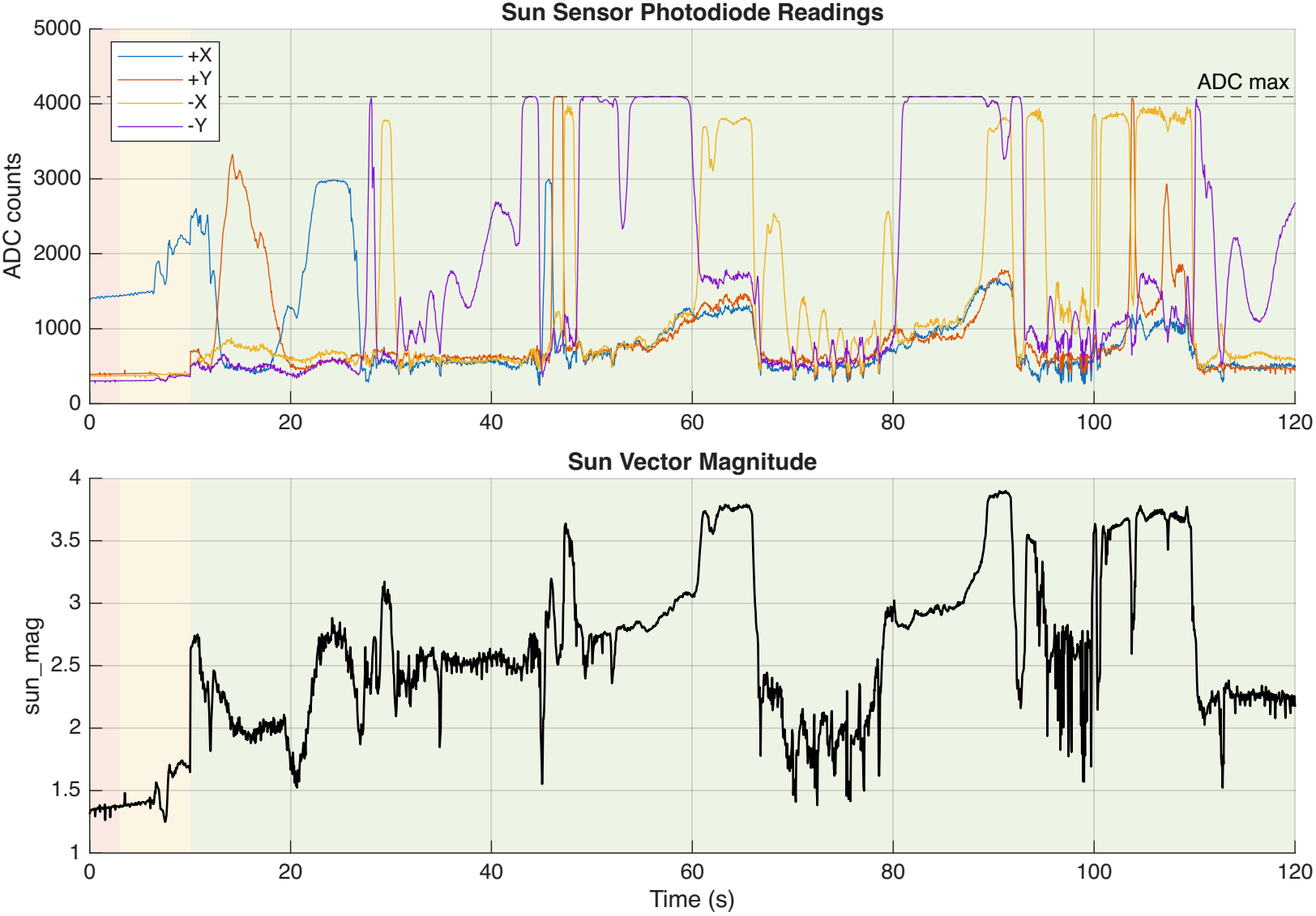
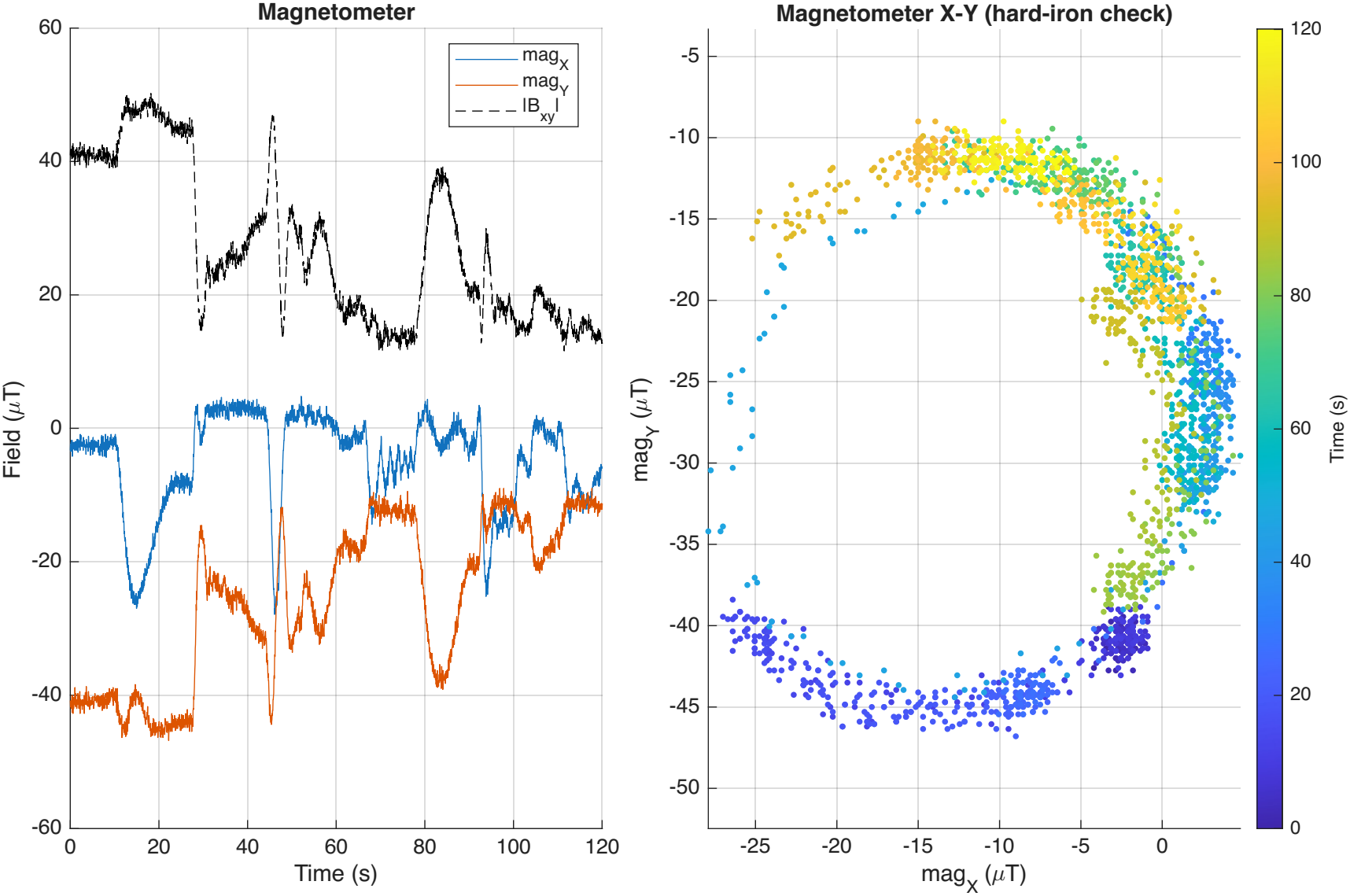
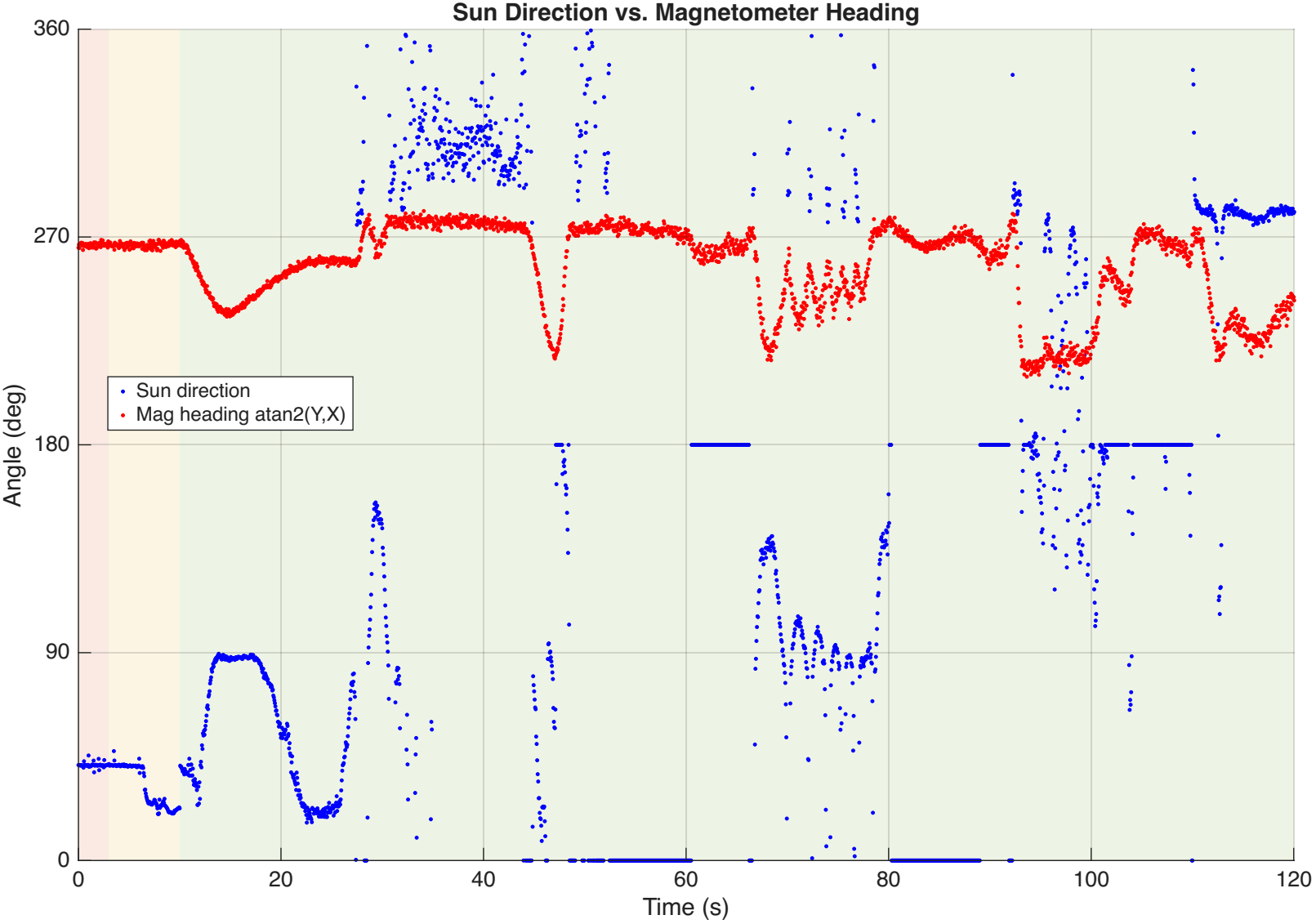

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

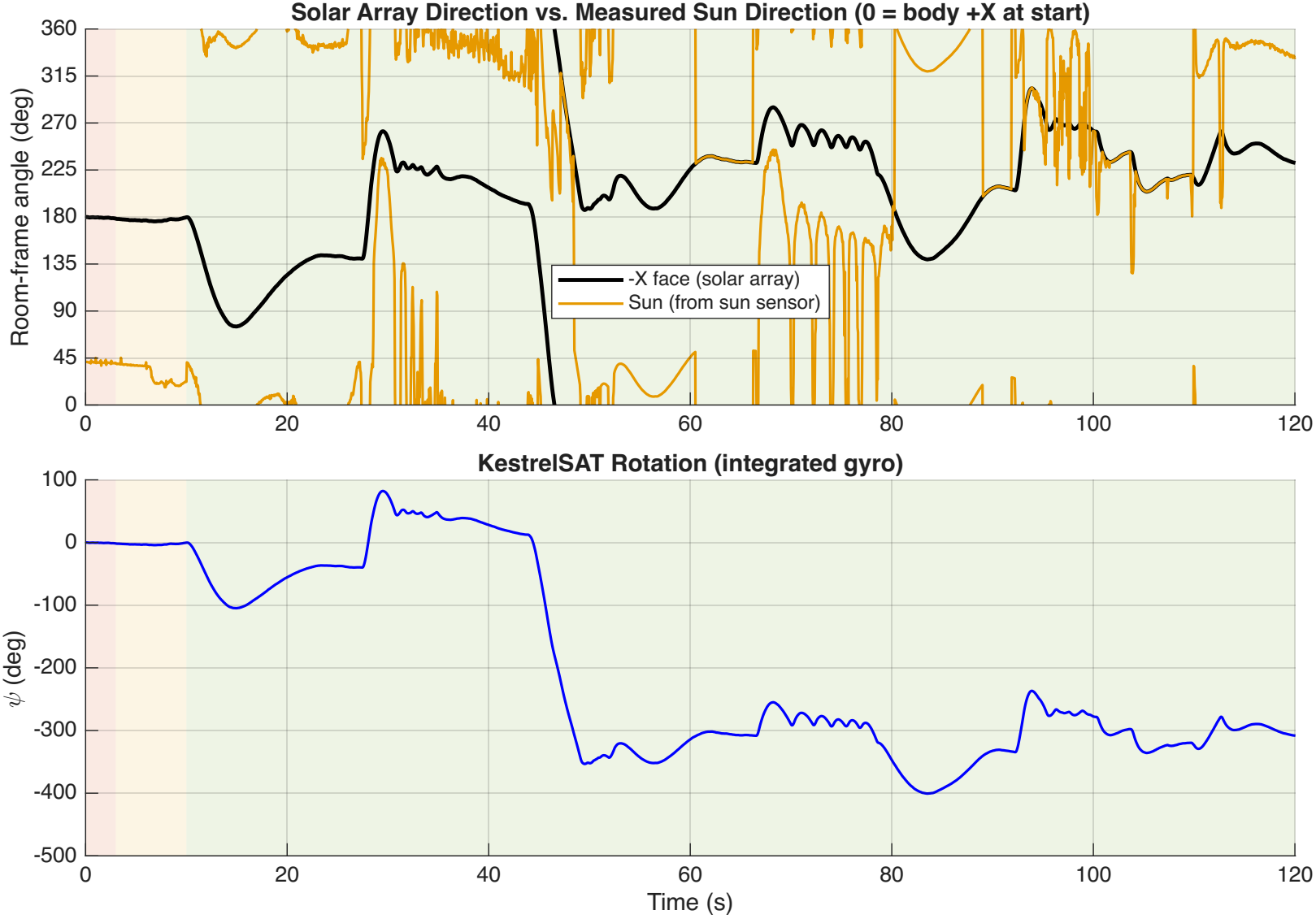

Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
150
100
50
0
-50
-100
-150
0
20
40
60
80
100
120
SPINUP
HOLD
TRACK

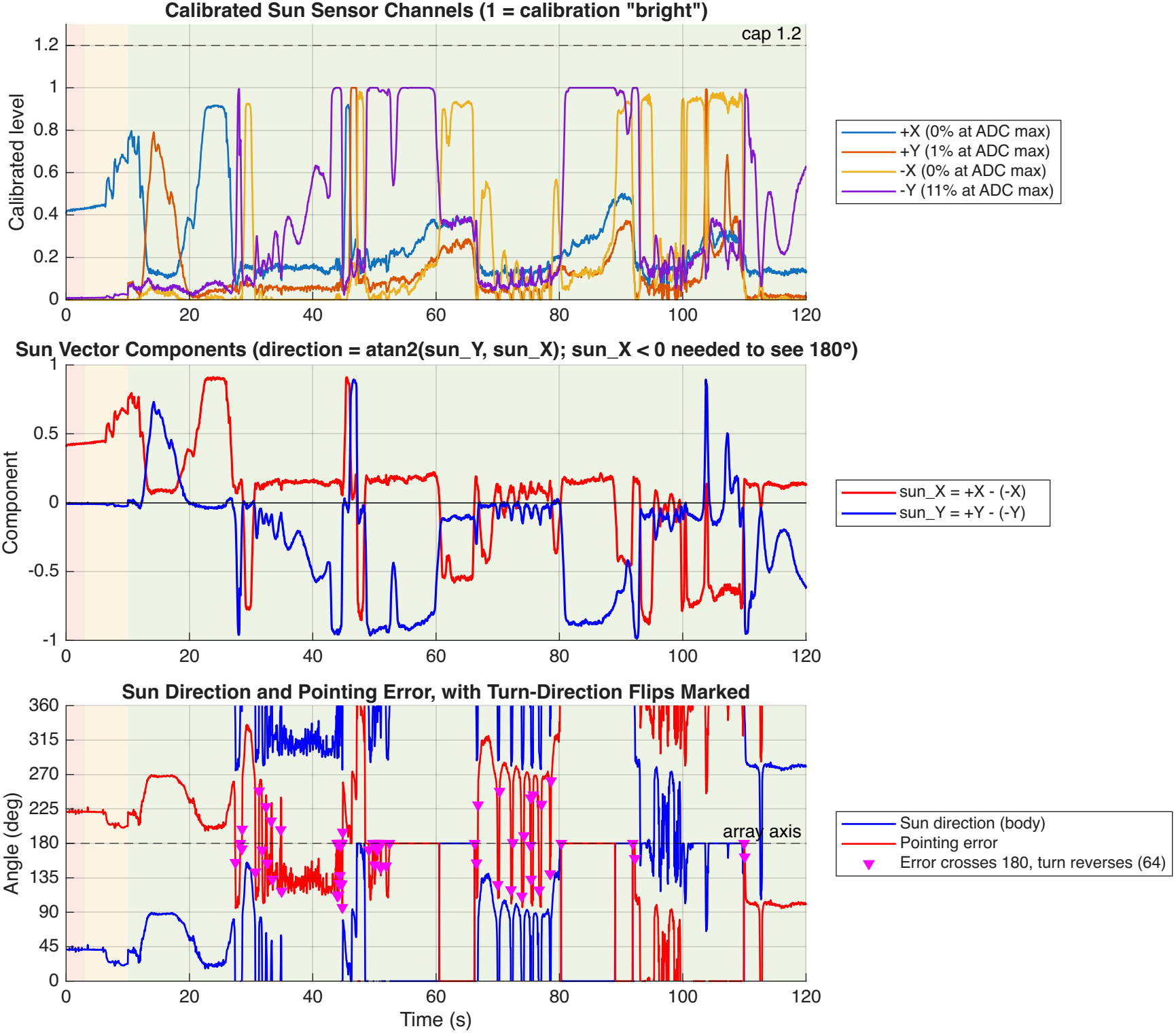

Reaction Wheel Speed
Commanded
Measured
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
120
Commanded Torque (magenta = wheel saturated)
Torque cmd
Saturated
Torque cmd (mN·m)
600
400
200
0
-200
-400
-600
Time (s)

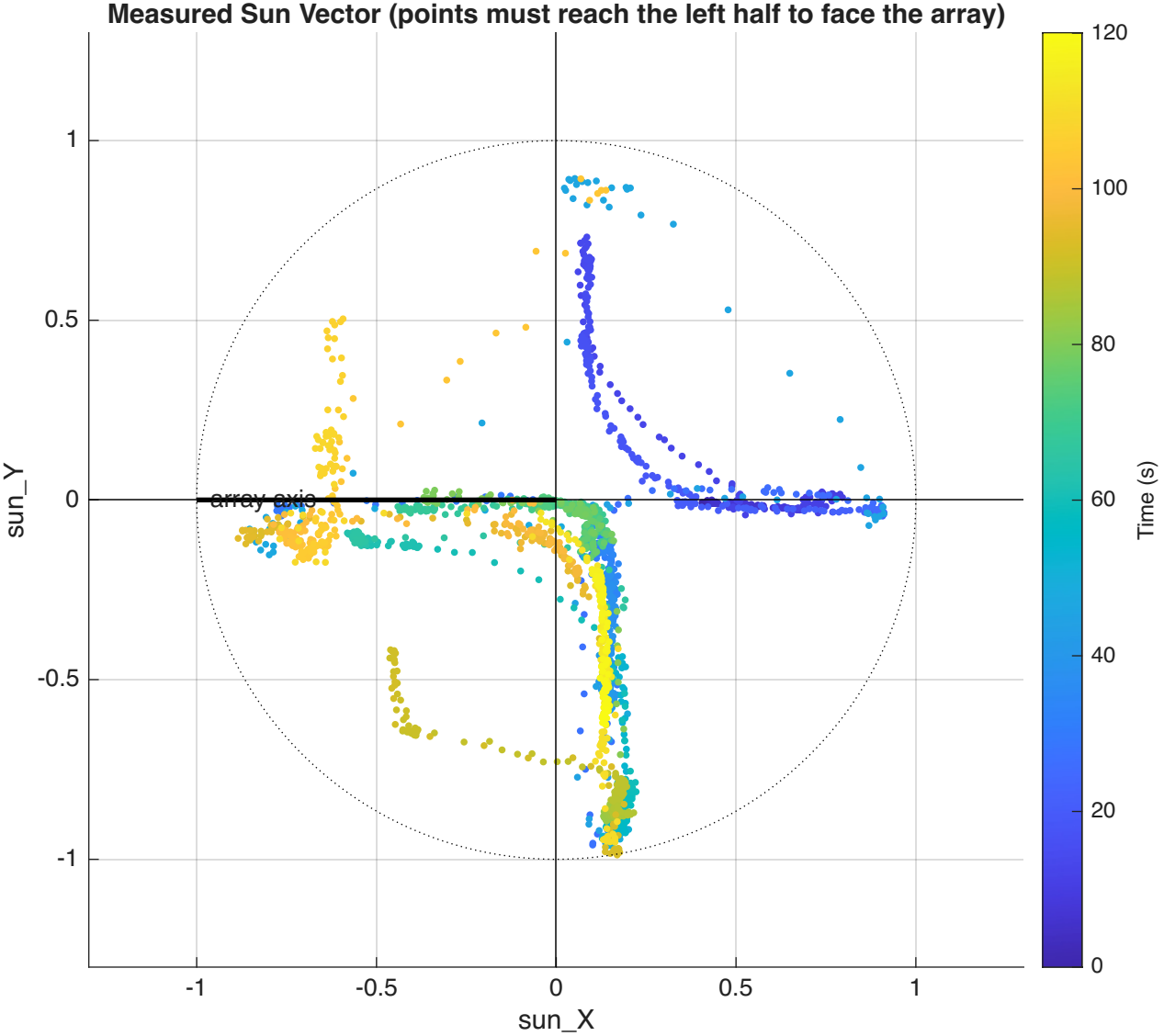

Sun Sensor Photodiode Readings
+X
+Y
-X
-Y
ADC max
ADC counts
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
Sun Vector Magnitude
sun_mag
4
3.5
3
2.5
2
1.5
1
Time (s)

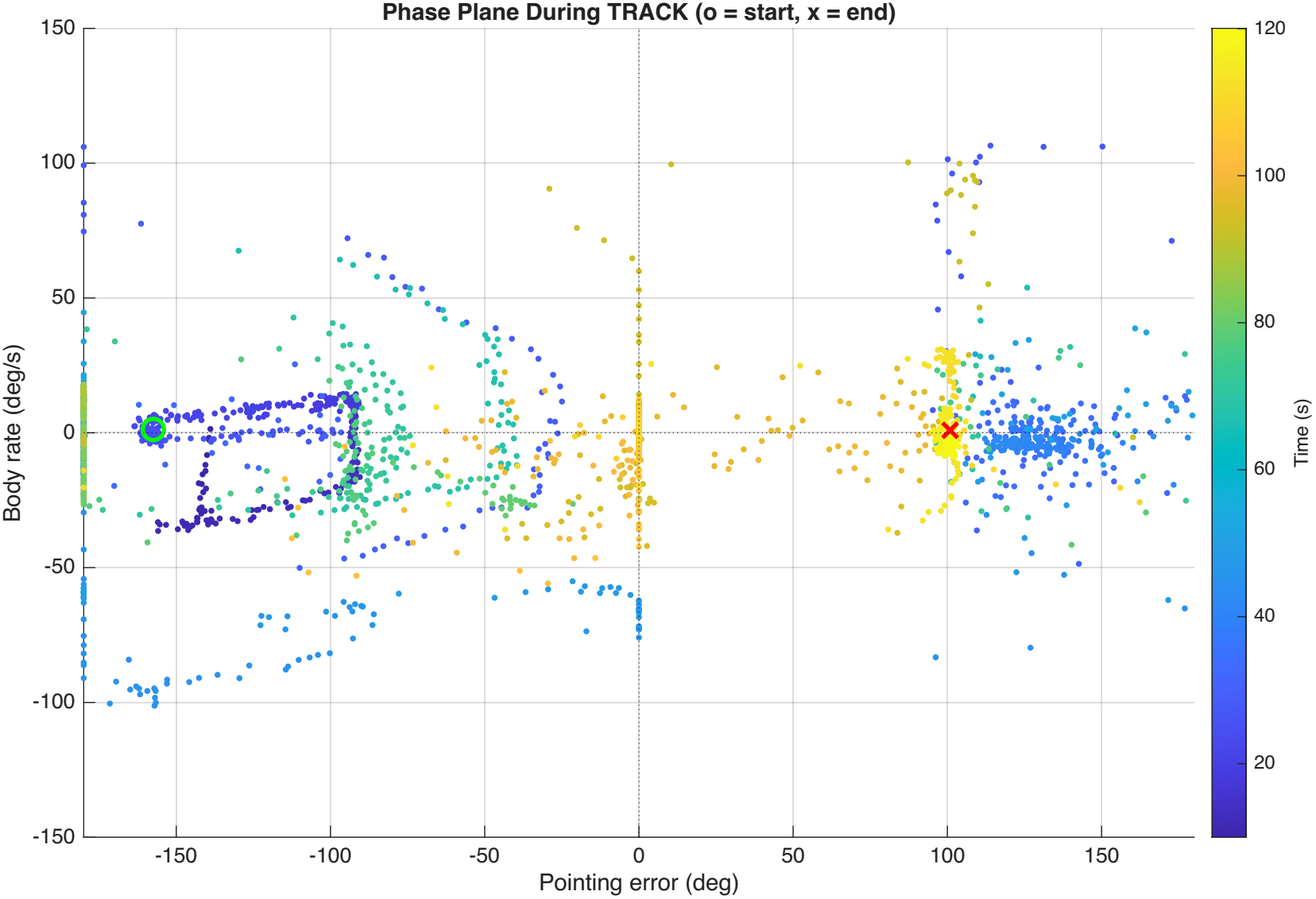

Magnetometer
magX
magY
|Bxy|
Field (μT)
Time (s)
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)

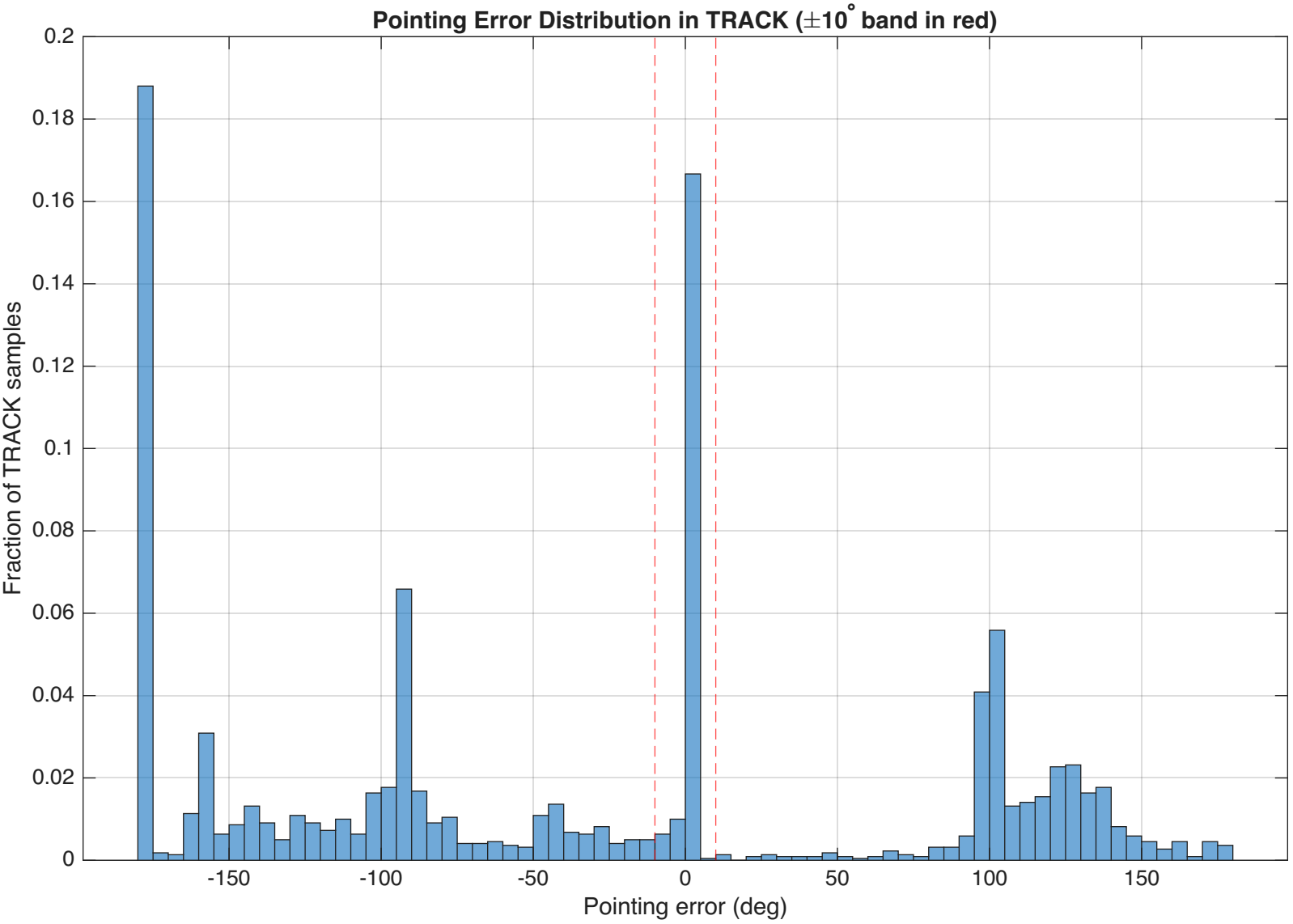

Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
Time (s)
360
270
180
90
0
0
20
40
60
80
100
120


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
20
40
60
80
100
120
-X face (solar array)
Sun (from sun sensor)
KestrelSAT Rotation (integrated gyro)
ψ (deg)
100
0
-100
-200
-300
-400
-500
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (1% at ADC max)
-X (0% at ADC max)
-Y (11% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Error crosses 180, turn reverses (64)
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
120
100
80
60
40
20
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
150
100
50
0
-50
-100
-150
-150
-100
-50
0
50
100
150
120
100
80
60
40
20


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-150
-100
-50
0
50
100
150